Problem-Solution Fit

Date	7 July 2026
Team ID	xxxxxx
Project Name	EduGenie Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:
Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience. Fill out the canvas in the respective boxes just like the original visual template.

1. CUSTOMER SEGMENT(S) [CS] Students, self-learners, and educators looking for structured, interactive learning support across various academic levels.	**6. CUSTOMER LIMITATIONS** EG. BUDGET, DEVICES **[CL]** Limited access to expensive cloud AI subscriptions and hardware constraints like resource-constrained personal computers (e.g., Mac M1).	**5. AVAILABLE SOLUTIONS** PROS & CONS **[AS]** Commercial cloud LLMs offer high accuracy but cost more, while traditional study portals provide static content without interactive quiz feedback.
2. PROBLEMS / PAINS + ITS FREQUENCY **[PR]**	**9. PROBLEM ROOT / CAUSE [RC]**	**7. BEHAVIOR** + ITS INTENSITY **[BE]**

Daily overload from complex study materials, missing self-evaluation tools, and fragmented, unclear domain progression paths.

Standard educational resources are static, unyielding, and non-adaptive, forcing learners to waste time manually parsing data.

High-intensity cramming sessions during exams mixed with steady, moderate-intensity study patterns for long-term professional upskilling.

3. TRIGGERS TO ACT [TR]

Impending exam dates, assignment deadlines, or immediate technical skill gaps requiring a clear learning strategy.

4. EMOTIONS

BEFORE / AFTER **[EM]**

BEFORE : Overwhelmed by heavy textbooks, highly anxious about exam preparation, and frustrated by confusing domain progression paths.

AFTER : Relieved by clean text summaries, confident in self-evaluation results, and motivated by clear step-by-step progress roadmaps.

10. YOUR SOLUTION [SL]

EduGenie, a hybrid educational web application built on FastAPI that optimizes local and cloud AI resources to provide fast, lightweight, and interactive student tutoring features.

8. CHANNELS OF BEHAVIOR [CH]

ONLINE

Connecting to high-performance cloud AI engines (Google Gemini 1.5 Pro) through web browsers for deep learning workflows.

OFFLINE

Processing local text utilities and basic summarization tasks via local models (LaMini-Flan-T5) without internet dependencies.